THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A raisin placed in water: 1

(a) swells up **(b)** dissolves **(c)** does not change **(d)** shrinks

2. The movement of water through a selectively permeable membrane from a region of higher water concentration to lower is: 1

(a) osmosis **(b)** diffusion of solute
(c) transpiration **(d)** evaporation

3. Plastids that give colour to flowers and fruits are: 1

(a) chloroplasts **(b)** chromoplasts **(c)** mitochondria **(d)** leucoplasts

4. A cell placed in an isotonic solution: 1

(a) bursts **(b)** swells
(c) shrinks **(d)** shows no net movement of water

5. The control centre of a eukaryotic cell is the: 1

(a) nucleus **(b)** vacuole **(c)** ribosome **(d)** lysosome

6. The colourless plastids that store starch are: 1

(a) lysosomes **(b)** chromoplasts **(c)** leucoplasts **(d)** chloroplasts

7. Cell division in which one parent cell gives two identical daughter cells is: 1

(a) fertilisation **(b)** mitosis **(c)** budding only **(d)** meiosis

8. The cell wall of plants is made of: 1

(a) fat **(b)** protein **(c)** cellulose **(d)** chitin

9. Large central vacuoles are present in: 1

(a) red blood cells **(b)** bacteria
(c) animal cells only **(d)** mature plant cells

10. Meiosis produces: 1

(a) gametes with half the chromosome number **(b)** identical cells
(c) no cells **(d)** twice the chromosomes

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two differences between plant and animal cells. 2

12. Why is the nucleus called the control centre of the cell? 2

13. State two functions of the cell membrane. 2

14. Why must a cell divide instead of growing indefinitely? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the function of the Golgi apparatus and the endoplasmic reticulum. 3

16. Give three evidences for the endosymbiotic origin of mitochondria and plastids. 3

17. State three differences between a prokaryotic and a eukaryotic cell. 3

OR

Name the cell organelles with the functions: mitochondria, ribosomes, lysosomes. 3

18. What is cell culture? State two uses. 3

OR

Why do animal cells lack a cell wall but plant cells have it? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the cell theory, its contributors and its exceptions or modern additions. 5

OR

Explain cell culture and tissue culture, with their uses in medicine and agriculture. 5

20. Explain the effect of different solutions on plant cells and animal cells (red blood cells) with plasmolysis and haemolysis. 5

OR

Explain the functions of the endoplasmic reticulum, Golgi apparatus and lysosomes in the cell as an assembly line. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND